

Parth Desai

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SUMMARY

Robotics Software Engineer with 2+ years of experience building end-to-end autonomy stacks for robotic systems and autonomous vehicles. Experienced across perception, localization, SLAM (frontend/backend), state estimation, motion planning, and multi-sensor fusion for real-time autonomy. Skilled in developing ROS2-based architectures, nonlinear optimization for localization and mapping, and deploying planning algorithms such as RRT*, BIT*, and MPPI. Adept at simulation-driven development using CARLA, Gazebo, and MuJoCo, and building reliable software through Dockerized workflows, CI/CD pipelines, and digital twin validation.

TECHNICAL SKILLS

Programming & Core Languages: C++, Python, Bash, MATLAB, Object-Oriented Design, Multithreading, Algorithm Optimization, Software Design Patterns, Memory Management, Parallel Computing (CUDA).

Robotics Frameworks & Middleware: ROS 2, DDS (FastDDS), DDS QoS, TF2/URDF Modeling, Nav2, MoveIt2, BehaviorTree.CPP, Launch Files, Parameter Server, Node Composition, RViz, Gazebo/IGN, Lifecycle Management.

Perception, Vision & AI: OpenCV, PCL, Open3D, PyTorch, YOLO, TensorRT, Object Detection & Tracking, 3D Point Cloud Processing, Image Segmentation, Neural Network Deployment on Embedded Systems.

Estimation, Localization & Sensor Fusion: EKF/UKF, Multi-Sensor Fusion, SLAM (frontend: scan matching / feature association, backend: Pose Graph Optimization), Mahalanobis/ IoU Gating, Odometry (Wheel, LiDAR, Visual, Inertial), Nonlinear Optimization (Ceres, GTSAM), Lie Groups (SO(3), SE(3)).

Planning & Control: OMPL (BIT*, RRT*), DWB/TEB, A*/D*, MPPI/MPC, PID Cascades, Pure Pursuit, Kinematics-Based Motion Control, Trajectory Optimization, Path Tracking, Obstacle Avoidance, Dynamic Replanning.

Simulation, Systems & DevOps: Linux, Git, Docker, CI/CD, Gazebo, CARLA, MuJoCo, Automated Testing, Performance Profiling.

PROFESSIONAL EXPERIENCE

Robotics Software Engineer, John Deere

Jan 2025 - Present | IA, USA

- Engineered a deep-learning perception pipeline using YOLO and PyTorch, enabling real-time object detection and crop classification on ARM-based embedded GPU platforms with ROS 2 integration.
- Designed LiDAR - camera fusion architecture using ESEKF on SE(3) manifolds with Mahalanobis gating, using GTSAM for backend factor graph optimization, improving 3D tracking accuracy and reliability under variable agricultural field conditions.
- Implemented MPC and PID cascade controllers for autonomous vehicle path tracking, enhancing control precision and reducing trajectory deviation by 18% across uneven terrain.
- Developed CARLA - ROS2 simulation environment integrated with MuJoCo for digital twin validation, accelerating perception and control testing cycles across virtual field scenarios.
- Automated perception benchmarking using Dockerized CI/CD pipelines with GitLab runners, streamlining performance validation, latency monitoring, and inference accuracy for continuous model deployment.

Robotics Software Engineer, Bosch Global Software Technologies

May 2022 - Jul 2023 | Pune, India

- Developed ROS2-based navigation and motion control modules integrating Nav2, BehaviorTree.CPP, and MoveIt2, enhancing autonomous vehicle maneuverability and obstacle avoidance accuracy by 30%.
- Implemented multi-sensor fusion pipeline combining LiDAR, IMU, and GPS using EKF/UKF, improving real-time localization stability in dynamic indoor and outdoor environments.
- Deployed ICP/NDT point cloud registration for high-precision AMR localization against pre-built static maps, ensuring robust navigation across variable lighting and terrain conditions.
- Built custom path planning algorithms using OMPL (global: RRT*, BIT*, local: MPPI) for route optimization, reducing trajectory computation time by 25% across heterogeneous autonomous platforms.
- Deployed Linux-based ROS2 nodes via Dockerized CI/CD workflows, integrating deterministic launches, automated logging, and ROS Bag validation to ensure reproducible robotic software performance.

PROJECTS

Bandwidth-Aware Edge Perception ([Github](#) | [Project](#)): Engineered a fault-tolerant ROS 2/FastDDS system on Jetson Orin Nano, with TensorRT-optimized YOLOv8. Implemented adaptive bandwidth throttling to maintain stability under 10% packet loss and a watchdog for <2s automated recovery.

ADAS – Lane Keeping & AEB System ([Github](#) | [Project](#)): Developed a ROS 2-based ADAS stack in CARLA featuring multi-modal LiDAR, radar, and camera fusion for robust environmental perception. Engineered Pure Pursuit and PID cascade controllers to handle dynamic traffic scenarios, achieving 95% Autonomous Emergency Braking (AEB) reliability and <0.22m lateral tracking error.

HELIOS – Search & Rescue Drone ([Github](#) | [Project](#)): Built a PX4-controlled autonomous UAV for disaster response, fusing LIO, VIO, and IMU via ESEKF with GTSAM-based factor graph optimization for drift-resistant, GPS-denied navigation. Integrated a YOLO vision pipeline to autonomously detect survivors and transmit real-time images with localized global coordinates.

EDUCATION

Master of Science, Iowa State University

Aug 2023 - Aug 2025 | IA, USA

Computer Science (GPA: 3.72)

Bachelors of Engineering, Savitribai Phule Pune University

Aug 2019 - May 2023 | Pune, India

Computer Science Engineering (GPA: 9.30)